

What Carries Deep-Noise Robustness in a Mamba-3 Block? Input-Independent Scans and Noise-Dependent Gating under Cross-Condition Shift

Yi Wang*

College of Engineering, Lishui University, Lishui 323000, Zhejiang, China

Abstract

The robustness of Mamba-style selective state-space models is often attributed to input-dependent selectivity. We test this with controlled ablations of one Mamba-3 block on bearing vibration data under cross-condition shift (XJTU-SY, one speed-and-load pair run in both directions, training and evaluation bearings disjoint), with matched-noise training and final-epoch macro-F1 under a fixed budget. All central contrasts were run with training and evaluation noise drawn independently. Making the scan’s dynamics and read/write coefficients input-independent does not remove the block’s advantage over a time-invariant S4D: on the original transition the fully frozen block leads by +21.3, +23.3, +20.3 pp at 0, -2 and -6 dB, and on the reverse transition by +27.5, +26.7, +21.1 pp. Removing the multiplicative output gate costs +5.4, +9.3, +32.6 pp in the fully frozen block, resolved only at -6 dB; only in this host and direction does the cost grow significantly with noise depth, and without its gate the block still exceeds S4D at 0 and -2 dB but not at -6 dB. Multiplicative gating exceeds an additive branch of identical size by +23.3 pp at -6 dB. On Paderborn bearings the gate-removal cost replicates when training and evaluation share bearings. Grafting a gate onto S4D, per-class results and Paderborn with disjoint bearings were run only under an earlier noise protocol and are descriptive. The evidence concerns one block, one shift pair per dataset and five seeds.

Keywords: state-space models, Mamba, selectivity, gating, robustness,

*Corresponding author

Email address: jefflsxy@lsu.edu.cn (Yi Wang)

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (12–28 pp between S4D and the frozen Mamba-3 blocks in the independent-noise results), the evaluation labels are identical across all cells of each transition (hash-verified), and the block can be dissected.

Failure to identify the source of an empirical gain is a recurring weakness of machine-learning scholarship [6]. Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. We take a regime in which a Mamba-style block shows a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it, in three steps with adjudication thresholds fixed before each grid was run:

1. **Freeze the scan.** The input-dependent parts of Δ , A , B , C are replaced by learned constants; in a second arm the trapezoidal weight and rotation angles are as well, so that the scan’s dynamics and read/write coefficients no longer depend on the input. The value stream and the gate input remain input-dependent (Section 3.2).
2. **Remove the gate.** On the frozen base the multiplicative SiLU gate is removed, with the other components and the budget unchanged.

35 **3. Replace or transplant the gate.** In place, the gate is replaced by
 36 an additive branch of identical size and initial weights; across hosts, it
 37 is grafted onto plain S4D with an additive control.

38 Steps 1 and 2 and the in-place replacement were tested with training
 39 and evaluation noise drawn independently (Section 4); the graft onto S4D,
 40 the per-class analysis, the stem and A ablations and the disjoint-bearing
 41 Paderborn setting were run only under an earlier protocol in which the two
 42 noise streams were shared by window index, and are reported in [Appendix](#)
 43 [D](#) as descriptive evidence (Table 2). The outcome is a conditional pattern.
 44 Making the scan input-independent does not remove the block’s advantage
 45 over S4D, on either transition (+21.3, +23.3, +20.3 pp and +27.5, +26.7,
 46 +21.1 pp for the fully frozen block). Removing the gate costs +9.4, +15.7,
 47 +27.0 pp in the partially frozen block on the original transition, resolved
 48 only at -6 dB, and 19.3, 22.7, 22.0 pp on the reverse transition, resolved at
 49 every level. Only in the fully frozen block on the original transition does
 50 the gate-removal cost grow significantly with noise depth: without its gate
 51 that block still exceeds S4D at 0 and -2 dB but not at -6 dB. An additive
 52 branch of identical size does not recover the gate’s effect at -6 dB in either
 53 host, although the pre-specified rescue test is not assessable. On Paderborn
 54 bearings the gate-removal cost replicates when training and evaluation share
 55 bearings.

56 **Contributions.** (i) A layered freezing experiment on a Mamba-3 block,
 57 down to a scan with input-independent dynamics and read/write coefficients
 58 (verified at the kernel interface), in which the advantage over S4D survives
 59 on both transitions of one dataset under independent training and evaluation
 60 noise. (ii) Same-host tests of the output gate (removal and in-place additive
 61 replacement) in a partially and a fully frozen block, with a descriptive analysis
 62 of how the gate’s contribution changes with noise depth. As supporting
 63 material, the study reports a coverage table by protocol, the correction of an
 64 initial noise-protocol defect, and a released pre-specification and review trail,
 65 including the incidents in which automated adjudication departed from the
 66 pre-specified criteria (Appendix C).

67 **2. Background and Related Work**

68 *2.1. Time-invariant and selective SSMs*

69 Diagonal structured SSMs (S4D) [7], a simplification of S4 [8] whose ini-
 70 tialisation derives from HiPPO [9], evolve a linear state with input-independent

parameters (A, B, C, Δ) , acting as a bank of learned linear filters. Related alternatives to attention [10] include S5, a multi-input multi-output linear SSM [11], the linear recurrent unit [12], and Hyena, which interleaves implicitly parametrised long convolutions with data-controlled gating [13]. Mamba-style blocks [1] make Δ, B, C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [14] (see also Mamba-2 [15]). The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [16], and on [17]. We treat these as *hypotheses to be tested*, not established attributions: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

Multiplicative gating is not specific to Mamba. It has been used in recurrent networks since the LSTM [18], in highway networks [19], in gated convolutional language models [20] and their transformer variants [21], and as channel-wise recalibration in convolutional networks [22]. The gate of a Mamba-style block applies the SiLU activation [23, 24] to a parallel branch and multiplies the result onto the scan output; this is the operation whose role the present paper isolates.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds S4D-based models superior to Mamba variants [4]; that study does not include additive noise or a shift of operating condition between training and evaluation, which is the regime examined here; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5], and copying [25], state-tracking [26] and associative-recall [27] studies delimit tasks on which fixed-state sequence models lag attention. In vision, where Mamba backbones have been adopted [28], it has been hypothesised that Mamba is not necessary for image classification [29]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [30, 31, 32, 33, 34], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D

108 matches or slightly exceeds the Mamba-style block on clean data). Whether
 109 the disagreement in the literature reflects a regime difference is not settled
 110 by a single testbed; what we can report is that, in the drift $\times$ noise regime
 111 studied here, the Mamba-style block’s advantage over S4D is not removed by
 112 freezing the classical selectivity terms.

113 2.3. Testbed rationale

114 We use bearing vibration diagnosis under cross-condition drift (speed
 115 and load both change) as the testbed, for three properties that matter to
 116 the attribution question rather than to the application: (a) the distribution
 117 shift is physically generated (a change of rotation speed and radial load be-
 118 tween recorded conditions), not simulated; (b) fault signatures are sparse
 119 transients—precisely the discontinuous structures selectivity theory predicts
 120 selective SSMs should excel at [16], which makes it a plausible arena for the
 121 selectivity hypothesis; and (c) the robustness gap between the Mamba-style
 122 block and S4D is large (12–28 pp in the independent-noise results), giving
 123 the attribution chain high signal. Bearing diagnosis with machine learning is
 124 reviewed in [35, 36]; the Case Western Reserve University data are the best-
 125 known benchmark [37] and have been used to study noise and load changes
 126 with convolutional networks [38], while open-source benchmark studies stress
 127 comparable protocols [39]. Operating-condition shift is usually addressed
 128 with domain adaptation such as adversarial training [40]; we use none, be-
 129 cause the question is which component of one block carries robustness, not
 130 how to close a domain gap. Additive noise at a fixed SNR is applied in the
 131 manner of corruption benchmarks [41]. The testbed is two-class (details in
 132 Section 3.1); the scope of all claims is bounded accordingly (Section 6).

133 3. Experimental Protocol

134 3.1. Testbed, arms, and budget

135 Vibration windows from the XJTU-SY run-to-failure dataset [42] are s-
 136 plit by operating condition: models train on one operating condition (37.5 Hz,
 137 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change
 138 (cross-condition), with additive white Gaussian noise applied at matched S-
 139 NR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The
 140 task is binary fault classification (inner-race vs. outer-race dominant class-
 141 es; class imbalance makes IR the hard class, Appendix D.10). All arms
 142 share an identical training budget (50 epochs, 5 seeds; the per-class grid

143 of [Appendix D.10](#) used 20 epochs), identical splits, and identical evalua-
 144 tion labels, verified by a hash of the label vector: every cell on the original
 145 transition carries `eval_sha = 6c20b367...` ([Appendix A](#)), recomputed inde-
 146 pendently during adjudication; the reverse transition and the two Paderborn
 147 settings carry their own label hashes (`93424834...`, `53d565d9...` disjoint,
 148 `addc1938...` shared). The hash certifies the label vector, not the input win-
 149 dows. In total the study comprises 865 cells across five pre-specified grids
 150 of the original design (baseline grid 475; frozen-selectivity 105; component
 151 ablation 175; graft 35; per-class 75), run under the earlier noise protocol, and
 152 follow-up grids add 645 cells: 210 under independent training and evaluation
 153 noise (Sections [4.1–4.3](#)) and 435 under the earlier protocol ([Appendix D](#));
 154 Table [2](#) gives the coverage. Training and evaluation bearings are disjoint in
 155 each direction: Bearing2_1, 2_2, 2_4, 2_5 at 37.5 Hz/11 kN and Bearing3_1,
 156 3_3, 3_4, 3_5 at 40 Hz/10 kN; the reverse transition swaps their roles.

157 **Composition of the data.** Training uses the bearings of the 37.5 Hz/11 kN
 158 condition whose failure mode is inner race or outer race (Bearing2_1 is la-
 159 belled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of
 160 the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5
 161 are OR). Bearings with cage or mixed failure modes are excluded. Windows
 162 are non-overlapping segments of 2048 samples of the horizontal channel taken
 163 from the fault phase, which is delimited by a kurtosis/RMS onset criterion
 164 with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR
 165 windows, the training set 4,624 OR and 624 IR windows. Two consequences
 166 matter for interpretation. The IR class is learned from a single training bear-
 167 ing, so class-level results may reflect bearing-specific generalisation. And the
 168 five seeds resample initialisation, batch order and noise but not bearings:
 169 there are four independent bearings on each side, and none of the intervals
 170 reported here captures bearing-level uncertainty. One follow-up grid (Sec-
 171 tion [4.2](#)) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the
 172 same eight bearings with their roles swapped, so it tests the other direction
 173 rather than independent bearings; its evaluation set holds 4,624 OR and
 174 624 IR windows, all IR windows coming from Bearing2_1, and its label hash
 175 is `93424834...`. The 35 Hz/12 kN condition has no inner-race bearing and
 176 cannot form a binary pair.

177 The principal arms are summarised in Table [1](#).

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Appendix D.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Appendix D.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Appendix D.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: scan dynamics and read/write coefficients input-independent (value stream and gate still input-dependent)	103,842
frozen-all-gate	frozen-all with the gate rows removed from the input projection (Section 4.1)	71,074
frozen-all+add	frozen-all with $y \odot \text{SiLU}(z)$ replaced by $y + \text{SiLU}(z)$; same parameters and initial weights	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Appendix D.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Appendix D.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$;	112,410

178 3.2. What the frozen arm freezes, and what the checks verify

179 In the Mamba-3 layer used here, one input projection emits the gate and
 180 value streams, the scan inputs B and C , the step-size and decay terms (Δ
 181 and A , through input-dependent additive terms), a trapezoidal weight and
 182 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 183 A , B and C and replaces them with learned constants broadcast over time
 184 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 185 rameter plays the role of a static log-decay, passed through the layer’s own
 186 heavy-tailed clamp). The removed projections are deleted, so no parameter
 187 or gradient can leak. **The trapezoidal weight and the rotation angles**
 188 **remain functions of the current input**, as do the stem and the output
 189 gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3
 190 formulation [14] the input-dependent rotation and trapezoidal weighting also
 191 enter the state update. [Appendix D.1](#) is accordingly a statement about the
 192 classical selectivity terms Δ , A , B , C only. Three further arms close that gap
 193 by the same device: *frozen-trap* replaces the trapezoidal weight by a learned
 194 per-head constant, *frozen-rot* replaces the rotation angles by learned con-
 195 stants (so the rotation becomes a learned position-only phase rather than a
 196 function of the current input), and *frozen-all* does both, so that the scan’s
 197 dynamics and read/write coefficients (Δ , A , B , C , trapezoidal weight, rota-
 198 tion angles) no longer depend on the input. The value stream x and the gate
 199 input z are still computed from the input, as they must be for the layer to
 200 carry any signal; the rotation becomes a learned position-dependent phase.
 201 In each case the corresponding `in_proj` output slice is deleted. [Appendix](#)
 202 [D.2](#) reports them. For these arms the construction check adds the decisive
 203 probe: the tensors handed to the scan are captured for two different inputs
 204 through the same module, and a tensor the arm claims to have frozen is
 205 required to be bitwise identical across them while a tensor left selective is
 206 required to differ (all three arms pass). No kernel modification is needed,
 207 because the scan interface accepts these tensors regardless of their origin.

208 Single-variable discipline was checked by key-level state-dictionary diffs
 209 between two models built under the same seed. For the frozen arm the
 210 diff verifies that every non-target tensor has the same name and shape as in
 211 BM3; for the component-ablation and graft arms it additionally compares the
 212 values of non-target tensors at initialisation. Every differing key is mapped
 213 to an entry in an architecture map. We do not claim byte-level identity of
 214 non-target tensors in the frozen arm, and trained weights naturally differ
 215 between arms; the checks concern how the arms were constructed.

216 3.3. Reporting rule

217 Every cell is trained for a fixed budget of 50 epochs with AdamW [43] and
 218 a cosine learning-rate schedule [44] that anneals to zero, and the macro-F1
 219 reported in the main text is that of the *final epoch*, a rule that uses no eval-
 220 uation data to choose a model. The original grids recorded instead the max-
 221 imum over epochs of macro-F1 on the evaluation split (the harness defines
 222 no separate validation split; the choice of model-selection rule can change
 223 conclusions in domain-generalisation benchmarks [45]), and the thresholds
 224 pre-specified for those grids were written against that rule. The arms of the
 225 original transition were therefore re-run on the adjudication band (0, −2,
 226 −6 dB) with the original training code, the same evaluation fingerprint and
 227 per-epoch evaluation (final-epoch re-run, earlier noise protocol; Appendix
 228 D.9). This final-epoch re-run and the independent-noise re-run of Section 4
 229 are different sets of cells with different coverage (Table 2). Best-epoch values
 230 are a sensitivity analysis. Levels outside the band (clean, +10, +6, −10 dB)
 231 and the per-class grid were not re-run; values there are best-epoch and aux-
 232 iliary. Because the final epoch of the same evaluation condition has been
 233 inspected repeatedly during this study, final-epoch values are not a fresh
 234 confirmatory test either.

235 3.4. Confound elimination

236 Before component attribution, three candidate confounds were examined
 237 on the same grid. **Capacity (earlier protocol):** S4D-wide ($1.75\times$ param-
 238 eters) tracks S4D (S4D-wide − S4D: −0.4, −1.0, +0.4 pp at 0, −2, −6 dB
 239 under final-epoch reporting, all intervals including zero), while BM3-frozen
 240 exceeds S4D-wide by +15.9, +20.8, +16.0 pp with intervals excluding zero
 241 at all three levels, so this particular increase in S4D’s parameter count does
 242 not explain the gap; the control does not address the change in channel mix-
 243 ing that gate removal also entails. **Normalisation:** both SSM arms use
 244 LayerNorm and contain no BatchNorm [46]; the arm that degrades under
 245 noise is the one *without* any batch statistics, which argues against Batch-
 246 Norm statistics mismatch as the explanation. Other normalisation inside
 247 the Mamba-3 layer was not varied. **Noise streams:** in the original harness
 248 the noise added to window i was drawn from a generator keyed on (seed, i),
 249 and training and evaluation used the same seed, so window i of the training
 250 set and window i of the evaluation set carried the same base noise vector.
 251 Per-index label agreement between the two sets is 0.549 on XJTU-SY, be-
 252 low the 0.628 expected by chance, which excludes only the simplest route

(memorising the training noise attached to each index); on Paderborn it is 0.999 (disjoint) and 1.000 (shared). All central contrasts were therefore re-run with the stream keyed on (seed, split, i), which makes training and evaluation noise independent while all arms with a given seed share one evaluation noise bank (Section 4.1); the reverse transition was re-run in the same way (Section 4.2). The component ablations of the stem and the state parameterisation, the grafts, the additive substitution on the reverse transition, the per-class grid and the disjoint-bearing Paderborn setting were not re-run; they carry the earlier protocol in their captions and are reported in Appendix D. In the arms that were re-run the means moved by at most 4 pp and no contrast changed direction; this is an observation about those arms, not a guarantee for the ones that were not re-run, and the earlier-protocol results are weighted accordingly. **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser carried by the original model contributes -0.1 pp pointwise (curves coincide); all attribution arms are therefore built on the pure cross-entropy variant.

3.5. Pre-specification and adjudication

Every adjudicated question in this study had numeric decision thresholds written to a task card and to disk before the corresponding grid was launched. This is internal pre-specification, not external registration: the supporting evidence is local file modification times and run-directory names (released, Appendix A), which are not tamper-evident and are not preserved by a repository checkout, so the trail lets readers inspect the order of events but cannot prove it. The exception is the follow-up grids from the width-matched graft onwards (Sections 4.1–4.3 and Appendix D.5–Appendix D.8): their thresholds, and for some of them the title decision that depends on their outcome, were committed to the public repository before the first cell was run (including the smoke check), which gives those grids an external time-stamp. Later questions (component ablation, graft) were specified after earlier grids had been seen, as is natural in a staged design. Adjudication was mechanical: an analysis pass applies the thresholds and emits a decision artifact; an independent reviewer pass re-computes aggregates, verifies fingerprints row-by-row, and must cite file paths and modification times for every check. The pre-specified thresholds were binding in practice: two attempted post-hoc criterion substitutions and one stale-artifact false verification were caught and re-adjudicated under the original rules — this is the one place where pre-specification actually bit. Appendix C details the incidents, the automated

290 executor/reviewer implementation, and the human adjudication gates; all
 291 scientific decisions — thresholds, rulings, and interpretations — were made
 292 by the authors and are timestamped. In the grids of the original design
 293 the adjudication criteria were effect-size thresholds, and paired intervals and
 294 tests were reported but not used as criteria. In the later grids (Section 4)
 295 the labels *holds*, *weakened* and *does not hold* are defined by whether the
 296 paired 95% interval excludes zero at two, one or none of three levels, so
 297 interval position is part of the adjudication there. A single repeat of one
 298 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1
 299 (non-deterministic kernel reductions; Appendix B); this is an observed repeat
 300 difference, not an estimated variability bound, and arm-level contrasts are
 301 judged against seed-level variability (2–9 pp).

302 4. Results

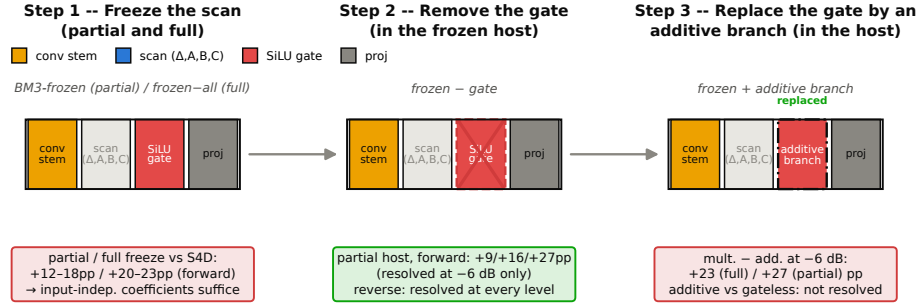
303 Throughout, “deep noise” denotes the pre-specified adjudication band
 304 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported only in the earlier-protocol appendix and
 305 excluded from adjudication (the time-invariant baseline is near chance there).
 306 All values are macro-F1 (%), mean over 5 seeds, at the final epoch of the
 307 50-epoch budget (Section 3.3). **The main text reports only results ob-**
 308 **tained with independent training and evaluation noise** (Sections 4.1–
 309 4.4). Results obtained while training and evaluation noise were shared by
 310 index (Section 3.4) are collected in Appendix D and summarised, with their
 311 evidence level, in Section 4.5. Table 2 shows which operation was run under
 312 which protocol on which host and transition. Figure 1 summarises the chain.

313 4.1. Original transition under independent noise

314 A pre-specified grid of 150 cells (thresholds, labels and writing conse-
 315 quences committed to the public repository before the first cell) re-ran the
 316 central arms with independent training and evaluation noise (Section 3.4):
 317 105 cells on XJTU-SY (the seven arms of Table 3, three levels, five seeds)
 318 and 45 on Paderborn shared bearings (three arms, Section 4.3); everything
 319 else is unchanged. Each contrast is labelled *holds* if at least two of the three
 320 levels have a positive mean with an interval excluding zero and none is re-
 321 versed, *weakened* if exactly one, *does not hold* if none. Relative to the earlier
 322 protocol the arm means moved by -3.4 to $+3.9$ pp over the eight arm-by-
 323 setting pairs that have an earlier-protocol counterpart (five XJTU-SY arms
 324 and three Paderborn arms); no contrast changed direction.

Table 2: Coverage of the study by protocol. I: run with independent training and evaluation noise (main text). E: run under the earlier protocol ([Appendix D](#)). -: not run. P: partially frozen host (BM3-frozen); F: fully frozen host. “Freeze” is the contrast with S4D (BM3-frozen – S4D for P, frozen–all – S4D for F); “gate removal” and “additive substitution” are applied inside the host. Graft, stem/ A removal and per-class recall have no fully frozen counterpart.

Operation	XJTU forward		XJTU reverse		PU shared		PU disjoint	
	P	F	P	F	P	F	P	F
Freeze (vs S4D)	I	I	I	I	I	–	E	E
Gate removal	I	I	I	–	I	–	E	–
Additive substitution	I	I	E	–	–	–	E	–
Graft onto S4D (with additive control)	E	–	E	–	–	–	–	–
Stem and A removal	E	–	–	–	–	–	–	–
Per-class recall	E	–	–	–	–	–	–	–



Conditional on this block, testbed and budget: input-independent scan coefficients suffice; gate cost grows with noise only in the forward fully frozen block.

Figure 1: The three-step chain. Steps 1 to 3 are tested under independent training and evaluation noise (Sections 4.1–4.4); step 3 is the in-host replacement of the multiplicative gate by an additive branch, on the partially and the fully frozen host. The graft of the gate onto S4D was run only under the earlier noise protocol and is an exploratory branch in [Appendix D](#). Numbers in the boxes are final-epoch differences in percentage points.

Table 3: Independent training and evaluation noise, original XJTU-SY transition (37.5 Hz/11 kN $\rightarrow$ 40 Hz/10 kN). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds. Lower block: paired differences with 95% t -intervals and the number of seeds (of 5) with a positive difference, and the pre-specified label. [‡]Added during review, not pre-specified.

Arm / contrast	0 dB	−2 dB	−6 dB	label
BM3-frozen	80.3	86.4	87.0	
frozen−all	89.5	91.5	88.9	
S4D	68.3	68.3	68.6	
frozen−gate	70.9	70.7	60.0	
frozen+add	71.3	71.3	60.1	
frozen−all−gate	84.1	82.2	56.3	
frozen−all+add	84.9	83.6	65.6	
frozen−all	+21.3	+23.3	+20.3	
− S4D	[14.6, 27.9] (5/5)	[20.2, 26.4] (5/5)	[17.9, 22.8] (5/5)	holds
BM3-frozen	+12.1	+18.1	+18.4	
− S4D	[0.9, 23.2] (5/5)	[9.8, 26.4] (5/5)	[12.7, 24.0] (5/5)	holds
BM3-frozen	+9.4	+15.7	+27.0	
− frozen−gate	[−0.5, 19.4] (5/5)	[−0.8, 32.1] (4/5)	[7.0, 46.9] (4/5)	weakened
frozen−all	+5.4	+9.3	+32.6	
− frozen−all−gate	[−3.8, 14.6] (3/5)	[−0.5, 19.1] (5/5)	[28.9, 36.3] (5/5)	weakened
frozen−all−gate	+15.9	+14.0	−12.3	
− S4D	[6.9, 24.8] (5/5)	[2.5, 25.4] (4/5)	[−17.5, −7.0] (0/5)	descriptive
BM3-frozen	+9.0	+15.1	+26.8	
− frozen+add	[−0.8, 18.9] (5/5)	[−0.9, 31.0] (4/5)	[6.8, 46.9] (4/5)	descriptive [‡]
frozen−all	+4.6	+7.9	+23.3	
− frozen−all+add	[−4.0, 13.2] (3/5)	[1.0, 14.8] (5/5)	[9.0, 37.5] (5/5)	descriptive [‡]
frozen+add	+0.4	+0.6	+0.1	
− frozen−gate	[−0.6, 1.4] (4/5)	[−0.1, 1.2] (5/5)	[−0.4, 0.7] (3/5)	descriptive
frozen−all+add	+0.8	+1.4	+9.3	
− frozen−all−gate	[−1.5, 3.1] (2/5)	[−1.9, 4.7] (2/5)	[−7.2, 25.8] (5/5)	descriptive

325 *Freezing.* Both freeze contrasts hold. The fully frozen block exceeds S4D
 326 by +21.3, +23.3, +20.3 pp, in every seed at every level; the partially frozen
 327 block exceeds it by +12.1, +18.1, +18.4 pp.

328 *Gate removal.* Removing the gate from the partially frozen block costs
 329 +9.4, +15.7, +27.0 pp; the contrast is *weakened* because only the -6 dB
 330 interval excludes zero. In the fully frozen block it costs +5.4, +9.3, +32.6 pp,
 331 again resolved only at -6 dB. Without its gate the fully frozen block still
 332 exceeds S4D by +15.9 and +14.0 pp at 0 and -2 dB (intervals excluding
 333 zero) and falls 12.3 pp below it at -6 dB (interval [-17.5, -7.0], all seeds).
 334 The gate is therefore not required for a positive advantage over S4D at 0 and
 335 -2 dB, but it is required at -6 dB. At the two milder levels the removal cost
 336 is +5.4 and +9.3 pp with intervals that include zero, so a contribution of the
 337 gate there is neither demonstrated nor excluded. The pre-specified condition
 338 for a combined attribution of the fully frozen block's advantage to the gate
 339 (gate removal holds and the additive substitution is not rescued) is not met,
 340 and we make no such claim.

341 *Multiplicative versus additive combination.* Replacing the gate by an ad-
 342 ditive branch of identical size and initial weights cannot be distinguished from
 343 removing it in either host (+0.4, +0.6, +0.1 pp and +0.8, +1.4, +9.3 pp; all
 344 intervals include zero), and the rescue ratios are small ($\rho = 0.029$, paired-seed
 345 interval [0.011, 0.059]; fully frozen host $\rho = 0.194$, [-0.002, 0.326]). Because
 346 gate removal is only weakened, the pre-specified rule labels the substitution
 347 test *not assessable*. This is a statement that no clear rescue was detected,
 348 not that the two arms are equivalent: in the fully frozen host the additive
 349 arm is +9.3 pp above the gateless arm at -6 dB (interval [-7.2, 25.8], 5/5
 350 seeds). The direct comparison is more informative: the multiplicative gate
 351 exceeds the additive replacement by +4.6, +7.9, +23.3 pp in the fully frozen
 352 host and by +9.0, +15.1, +26.8 pp in the partially frozen host, with intervals
 353 excluding zero at -2 and -6 dB in the former and at -6 dB in the latter
 354 (Table 3, descriptive). The ratio ρ is a conditional summary: among the
 355 3125 paired seed resamples, at least one noise level was excluded because its
 356 denominator fell below 2 pp (as pre-specified) in 16.2% (507) of the resamples
 357 for the fully frozen host and in 0.5% (16) for the partially frozen host; no
 358 resample excluded all three levels (the repository field `paired_frac_dropped`
 359 is this second statistic, 0 for both).

360 4.2. Reverse transition under independent noise

361 A second pre-specified grid (60 cells, four arms; thresholds committed
362 before the first cell) repeated the freeze and gate-removal contrasts on the
363 reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN, with the same bearings in
364 swapped roles, under independent noise. Relative to the earlier protocol the
365 arm means moved by -1.8 to $+1.7$ pp. This direction is harder for the time-
366 invariant model: on its evaluation set a majority-class predictor scores 46.8%
367 macro-F1 and a uniformly random one 41.5%, and S4D sits close to those
368 levels.

Table 4: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN), independent training and evaluation noise. Macro-F1 (%), mean over 5 seeds, best-epoch / final-epoch; paired differences under final-epoch reporting with 95% t -intervals, seeds positive and label. Fully frozen gate removal and additive substitution were not run in this direction under independent noise (Table 2).

Arm (best / final)	0 dB	-2 dB	-6 dB	label
frozen-all	88.6 / 77.4	88.1 / 77.6	81.1 / 73.3	
BM3-frozen	85.5 / 75.4	86.7 / 78.4	78.5 / 72.2	
frozen-gate	64.0 / 56.1	62.5 / 55.7	55.1 / 50.1	
S4D	61.2 / 49.9	60.2 / 50.9	55.7 / 52.2	
frozen-all	+27.5	+26.7	+21.1	
- S4D	[21.3, 33.8] (5/5)	[19.5, 33.9] (5/5)	[16.2, 26.0] (5/5)	holds
BM3-frozen	+25.5	+27.5	+20.0	
- S4D	[13.9, 37.1] (5/5)	[18.2, 36.9] (5/5)	[8.0, 32.0] (5/5)	holds
BM3-frozen	+19.3	+22.7	+22.0	
- frozen-gate	[2.9, 35.7] (5/5)	[4.1, 41.4] (5/5)	[3.2, 40.9] (5/5)	holds
frozen-gate	+6.2	+4.8	-2.0	
- S4D	[-15.6, 28.0] (2/5)	[-17.5, 27.1] (2/5)	[-19.6, 15.5] (1/5)	descriptive

369 The fully frozen block exceeds S4D by +27.5, +26.7, +21.1 pp and the
370 partially frozen block by +25.5, +27.5, +20.0 pp; removing the gate from the
371 partially frozen block costs 19.3, 22.7, 22.0 pp, with intervals excluding zero
372 and 5/5 seeds at every level. All three contrasts hold and none is reversed.
373 In this direction gate removal is thus resolved at every level for the partially
374 frozen host, whereas on the original transition it is resolved at -6 dB only;
375 the two directions are reported side by side and not merged. The gateless
376 arm cannot be distinguished from S4D (+6.2, +4.8, -2.0 pp). The contrasts
377 with S4D rest on a baseline close to the trivial floor. The reverse direction
378 shares the eight bearings of the forward one with swapped roles and has a

single inner-race bearing in its evaluation set, so it tests direction rather than new bearings.

4.3. Paderborn with shared bearings

On the Paderborn real-damage bearings (1500 $\rightarrow$ 900 rpm), a pre-specified 45-cell grid trained and evaluated on the same four bearings (KA04, KA16, KI04, KI14; different recordings) under independent noise (Table 5). Gate removal costs +7.6, +6.5, +4.4 pp with intervals excluding zero at all three levels (*holds*); the partially frozen block exceeds S4D by +5.6, +6.5, +8.3 pp and is resolved at -6 dB only (*weakened*). The fully frozen block, the additive arm and the graft were not run in this setting. The companion setting with bearings disjoint from the training set was run only under the earlier protocol, where per-index label agreement between training and evaluation windows is 0.999; it is reported in Appendix D and is not an independent-noise test. With two bearings per class a bearing’s identity determines its label in the shared setting, so a model can succeed partly by recognising bearing-specific signatures (a form of shortcut learning [47]); XJTU-SY, by contrast, evaluates on bearings disjoint from the training ones in each direction.

Table 5: Independent training and evaluation noise, Paderborn shared bearings (KA04, KA16, KI04, KI14; 1500 $\rightarrow$ 900 rpm). Paired differences under final-epoch reporting with 95% t -intervals, seeds positive and pre-specified label. Means: BM3-frozen 81.5 / 83.8 / 85.9, S4D 75.9 / 77.3 / 77.6, frozen-gate 73.8 / 77.3 / 81.5%.

Contrast	0 dB	-2 dB	-6 dB	label
BM3-frozen	+5.6	+6.5	+8.3	
– S4D	$[-4.4, 15.5]$ (4/5)	$[-2.9, 15.9]$ (4/5)	$[2.7, 13.8]$ (5/5)	weakened
BM3-frozen	+7.6	+6.5	+4.4	
– frozen-gate	$[2.7, 12.6]$ (5/5)	$[1.7, 11.3]$ (4/5)	$[3.2, 5.6]$ (5/5)	holds

4.4. Does the gate’s contribution grow with noise depth?

A contrast that is resolved at one level and not at another does not by itself show that the effect differs between levels. We therefore computed, with seeds paired, the difference between the gate-removal cost at a deeper level and at 0 dB (Table 6). This analysis was added during review and is descriptive.

Only the fully frozen host on the forward transition shows a change whose interval excludes zero at -6 dB (+27.2 pp, 5/5 seeds). The partially frozen

Table 6: Change of the gate-removal cost (frozen host – its gateless counterpart, percentage points) relative to 0 dB: mean, 95% paired t -interval and seeds (of 5) with a positive change. Independent noise, final epoch. Descriptive; not pre-specified; the eight differences are not corrected for multiple comparisons.

Host and transition	−2 vs. 0 dB	−6 vs. 0 dB
Forward, fully frozen	+3.9 [0.5, 7.3] (4/5)	+27.2 [19.2, 35.2] (5/5)
Forward, partially frozen	+6.2 [−6.3, 18.7] (4/5)	+17.5 [−1.2, 36.2] (4/5)
Reverse, partially frozen	+3.4 [−1.6, 8.5] (4/5)	+2.7 [−15.7, 21.1] (2/5)
PU shared, partially frozen	−1.1 [−3.1, 0.9] (1/5)	−3.2 [−8.4, 1.9] (1/5)

404 host on the forward transition has a positive point estimate with an interval
405 that includes zero; on the reverse transition the change is near zero although
406 the cost itself is large at every level; for Paderborn shared bearings the point
407 estimate is negative. The −2 versus 0 dB change for that block has a lower
408 interval bound close to zero and is not treated as a separate finding. The
409 statement that the gate’s contribution grows with noise depth is therefore
410 supported only by the fully frozen block on the forward transition, and we
411 restrict it to that case throughout; for the other settings the data are com-
412 patible with a depth-independent cost.

413 4.5. Evidence from the earlier noise protocol

414 Several results were obtained only under the earlier protocol (Table 2,
415 [Appendix D](#)). We list them with the weight we give them; none carries an
416 abstract-level claim by itself.

- 417 • *Graft onto S4D* ([Appendix D.4–Appendix D.5](#)). A gate branch of the
418 original width recovers $R = 0.196$ of the gap between S4D and BM3-
419 frozen; one rebuilt in the donor’s width recovers 0.296 (paired-seed
420 interval 0.14, 0.72), against 0.408 for an additive branch with identical
421 parameters. On the original transition the gate did not recover more
422 than the additive branch; on the reverse transition it recovered some-
423 what more ($\Delta R = 0.153$, paired-seed interval 0.01, 0.34). Descriptive.
- 424 • *Additive substitution on the reverse transition* ([Appendix D.6](#)): not
425 rescued by the pre-specified rule; not re-run under independent noise.

- 426 • *Stem and A removal* ([Appendix D.3](#)): no resolved effect of removing
427 the heavy-tailed A parameterisation; removing the stem improves
428 macro-F1 at 0 and -2 dB (intervals excluding zero) and costs 5.9 pp at
429 -6 dB (interval including zero).
- 430 • *Per-class recall* ([Appendix D.10](#), 20-epoch grid): gate-related differ-
431 ences concentrate on the minority inner-race class.
- 432 • *Paderborn with disjoint bearings* ([Appendix D.8](#)): none of the pre-
433 specified contrasts replicates and no architecture transfers well (47.7–
434 56.9%).
- 435 • *Reporting rule* ([Appendix D.9](#)): best-epoch values from the original
436 grids are optimistic by 0.6–10.6 pp relative to the final epoch; orderings
437 are unchanged.

438 5. Discussion

439 **What can be credited?** Within this Mamba-3 block, on this testbed
440 and with noise-aware training, four statements are supported by independent-
441 noise evidence. (i) Making the scan’s dynamics and read/write coefficients
442 input-independent did not remove the block’s advantage over S4D on either
443 transition of one dataset (the independent-noise grids contain no unfrozen
444 BM3 arm, so they do not show that freezing preserves all of the unfrozen
445 block’s performance): S4D trails the frozen block by 12–28 pp. This does not
446 show that input-dependence is useless: it shows that, in this block and un-
447 der this budget, learned constants suffice to keep the advantage, and the two
448 readings (“input-dependence contributes nothing here” and “what it con-
449 tributes is recovered by constants”) lead to the same attribution. (ii) The
450 gate-removal cost is large in the partially frozen block (resolved at -6 dB on
451 the original transition and at every level on the reverse one) and in the fully
452 frozen block (resolved at -6 dB). (iii) Without its gate the fully frozen block
453 still exceeds S4D at 0 and -2 dB, so the gate is not required for a positive
454 advantage there; the removal cost at those levels (+5.4 and +9.3 pp) has
455 intervals that include zero, so a contribution is neither shown nor excluded.
456 At -6 dB the gateless block falls below S4D. (iv) A multiplicative gate ex-
457 ceeds an additive branch of identical size and initial weights by +23.3 pp at
458 -6 dB in the fully frozen block and by +26.8 pp in the partially frozen block,
459 whereas the additive arm cannot be distinguished from the gateless one; the

460 pre-specified rescue test is nevertheless not assessable and equivalence is not
461 claimed.

462 **Noise-dependent gating.** Only one setting shows a gate contribution
463 that grows significantly with noise depth: the fully frozen block on the orig-
464 inal transition (change of the removal cost between 0 and -6 dB $+27.2$ pp,
465 interval excluding zero; Section 4.4). For the partially frozen block, the re-
466 verse transition and Paderborn shared bearings the data are compatible with
467 a depth-independent cost, and the statement in the title is restricted to the
468 first case. The depth analysis was added during review and is descriptive.

469 **Relation to earlier work.** Representability results for selectivity [16]
470 are not contradicted by a single ablation on one testbed. The S4D-superiority
471 finding [4] was obtained on clean classification benchmarks, without matched
472 noise or a shift of operating condition, so it does not speak to the regime
473 studied here. Two points of contact are nevertheless consistent with it: parity
474 between S4D and the Mamba-style arms on clean data (Appendix D), and
475 the finding that input-dependent selectivity is not needed for the block’s
476 advantage. What this paper adds is specific to the matched-noise, cross-
477 condition regime studied here, not a contradiction of the clean-benchmark
478 result.

479 **Implications for backbone selection and for transplant practice.**
480 For practitioners choosing backbones for noisy, drifting signal regimes, the
481 results suggest judging the block as a composition. Whether a gate added
482 to a different host reproduces its effect was tested only under the earlier
483 noise protocol (Appendix D.5): a gate built in the donor’s shape recovered
484 no more than an additive branch of the same size on the original transition
485 and somewhat more on the reverse one, and neither design was tuned; we
486 treat this as descriptive. A hypothesis for future work follows: a recomposed
487 block guided by the component map (no conv stem, gate retained, S4D-style
488 A) may match or exceed the original on this testbed; we leave it untested,
489 because post-hoc recomposition after seeing the map would not carry the
490 evidential status of the pre-specified chain.

491 **Why might gating be composition-bound?** We refrain from a mech-
492 anistic claim beyond the data, but note three compatible observations. First,
493 the gate-related differences concentrate on the minority, lowest-recall class
494 (Appendix D.10), and matched-noise training lifts that class *only when the*
495 *gate is present*. One hypothesis, which we did not test, is that the gate is the
496 component through which matched-noise training improves hard-class recall,
497 i.e., that it exploits noise-aware training rather than acting as a test-time de-

noiser; we measured neither classification margins nor gate activations, and the reading is in any case bounded by the matched-SNR scope (Section 6). Second, the multiplicative form admits a per-sample suppression interpretation, but no gate statistics were recorded, so this remains a hypothesis rather than an observation. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the weakest of the three observations. The gateless–S4D comparison depends on host and protocol: in the fully frozen block under independent noise the gateless arm is above S4D at 0 and -2 dB and below it at -6 dB (Table 3), whereas for the partially frozen block under the earlier protocol it is not resolved against S4D with five seeds (Appendix D.9). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed and datasets.** *Testbed scope.* All claims are bounded to bearing vibration from one pair of operating conditions of one dataset (37.5 Hz/11 kN and 40 Hz/10 kN; speed and load both change), binary classification, GPU execution (no edge/real-time claims). Within each direction the training and evaluation bearings are disjoint; the reverse direction uses the same eight bearings with their roles swapped, has a single IR bearing in its evaluation set (the forward direction a single IR bearing in its training set) and so tests direction rather than new bearings; the third condition of the dataset has no IR bearing. The seeds do not resample bearings (Section 3.1), so the intervals carry no bearing-level uncertainty. Further independent datasets and multi-class settings are future work. *Second dataset.* On Paderborn, the gate-removal cost replicates when training and evaluation share bearings under independent noise, and the partially frozen block’s advantage over S4D is resolved at -6 dB only. With bearings disjoint from the training set (earlier protocol only; per-index label agreement 0.999) nothing replicates and every architecture, S4D included, transferred poorly (47.7–56.9% against a balanced-chance level of about 50%). The two Paderborn settings differ in more than bearing overlap, so they do

not identify why transfer fails. The fully frozen block was run only in the disjoint setting. *Matched-SNR scope.* Training and evaluation share the same injected SNR throughout (Section 3.1); “robustness” in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate’s benefit may operate by exploiting matched-noise training (consistent with Appendix D.10) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ. *Noise-regime coverage.* Under strong coloured noise at -10 dB a TCN baseline slightly overtakes the Mamba-style block, and -10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.

2. **Noise protocol and coverage.** *Noise protocol.* All grids not marked in Table 2 as independent-noise used training and evaluation noise shared by index. Where both protocols exist the arm means differ by at most 4 pp and no contrast changed direction, which is not a guarantee for the arms that were not re-run. *Coverage by protocol.* Table 2 states which operations were run under independent noise. Not run under independent noise: the graft onto S4D, the stem and A ablations, additive substitution on the reverse transition, the per-class grid and the disjoint-bearing Paderborn setting. Not run at all: gate removal and additive substitution in the fully frozen block on the reverse transition, and the fully frozen block, additive arm and graft on Paderborn shared bearings. *Transplant results are earlier-protocol and budget-conditional.* The graft onto S4D was run only under the earlier noise protocol, with a matched 50-epoch budget; a longer or graft-tuned budget might recover more, and we make no claim beyond *this gate branch, in this host, under a matched training budget* (Appendix D.4).

3. **What the interventions can and cannot show.** *Scope of the freezing intervention.* The first freeze arm removes the input-dependence of Δ , A , B , C only; the fully input-independent scan is a separate grid that was specified after the earlier grids had been seen. No equivalence margin was pre-specified, so “no measurable cost” means the point estimates favour the frozen arms with intervals including zero, not equivalence; the independent-noise grids contain no unfrozen BM3 arm, so they do not show that freezing preserves the unfrozen block’s performance. The value stream and the gate input remain input-dependent, and freezing is implemented inside this Mamba-3 layer; a conclusion

about input-dependence in general would need other blocks and tasks. *Gate interventions.* Under independent noise, gate removal is resolved only at -6 dB in the partially and the fully frozen block on the original transition, and without its gate the fully frozen block still exceeds S4D at 0 and -2 dB; the combined statement that the fully frozen block’s advantage is carried by the gate is therefore not made. Where the gate-removal cost is resolved at every level (partially frozen block, reverse transition) it was not tested in the fully frozen block. The additive substitution could not be distinguished from gate removal, but the multiplicative-additive differences are positive at -6 dB; with gate removal only weakened the pre-specified rescue rule is not assessable, and neither equivalence nor zero rescue is claimed. The rescue ratio is a conditional summary (levels with a denominator below 2 pp are excluded in some resamples). *Attribution level.* Credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed. *Conv-stem dual reading.* Its clean-data cost admits both representation-mistransfer and optimisation-difficulty readings, reported in place ([Appendix D.3](#)). *Construction checks.* For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.

4. **Statistics, reporting and pre-specification.** *Sample size.* Every cell mean uses five seeds; run-to-run variance is a known confounder of benchmark conclusions [48, 49]. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem and A parameterisation; gateless versus S4D in the partially frozen block under the earlier protocol) are not resolved at this size. For the original-design grids conclusions rest on effect sizes that clear the pre-specified thresholds; for the later grids they rest on paired intervals as defined in Section 3.5. Seeds resample initialisation, batch order and noise (evaluation noise is one bank per seed), not bearings. *Reporting rule.* Reported values are final-epoch macro-F1 under a fixed 50-epoch budget. The best-epoch rule of the original grids, for which the early thresholds were pre-specified, is optimistic by 0.6–10.6 pp ([Appendix D.9](#)). The final-epoch re-run of the original-design arms (earlier noise protocol) and the independent-noise re-run cover different sets of cells (Table 2); levels outside the adjudication band and the per-class grid were not re-run. No source-domain

validation rule was available, and because the final epoch of the same evaluation condition has been inspected repeatedly, final-epoch values are not a fresh confirmatory test. *Strength of the pre-specification evidence.* Thresholds were pre-specified internally; for the follow-up grids from the width-matched graft onwards the thresholds were also committed to a public repository before the first cell, for the earlier grids the evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The noise-depth analysis (Section 4.4) and the multiplicative-additive differences were added during review and are not pre-specified. The process-level repeat check is a single observation.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled shift $\times$ noise testbed with matched-noise training and independent training and evaluation noise, making the scan’s dynamics and read/write coefficients input-independent did not remove the block’s advantage over a time-invariant S4D: the fully frozen block leads by +21.3, +23.3, +20.3 pp on the original transition and by +27.5, +26.7, +21.1 pp on the reverse one. Removing the multiplicative output gate causes a substantial loss at -6 dB, and this cost is resolved only at that level on the original transition (at every level for the partially frozen block on the reverse one; the ranking of the gate against the stem and A ablations comes from the earlier protocol and is descriptive), and without its gate the fully frozen block still exceeds S4D at 0 and -2 dB. Only in the fully frozen block on the original transition does the gate’s contribution grow significantly with noise depth; in the other settings tested the data are compatible with a depth-independent cost. A multiplicative gate exceeds an additive branch of identical size at -6 dB in both hosts, although the pre-specified rescue test is not assessable. On Paderborn the gate-removal cost replicates when bearings are shared between training and evaluation. Grafting a gate onto S4D, per-class results, the stem and A ablations and the disjoint-bearing Paderborn setting were run only under the earlier noise protocol and are descriptive. The evidence rests on one block, one shift pair

per dataset, five seeds and final-epoch reporting under a fixed budget. The independent-noise grids contain no unfrozen BM3 arm, so they establish that the frozen block beats S4D, not that freezing preserves the unfrozen block’s performance. We offer the chain—freeze, remove, replace—as a protocol for turning architecture comparisons into component-level tests, with two cautions from this study: interventions must be matched in width and capacity, and the noise streams of training and evaluation must be independent.

CRediT authorship contribution statement

Yi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used LLM-based agents to execute and cross-check pre-specified analysis and adjudication passes as described in Appendix C, and to assist with manuscript drafting and figure generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. All numeric thresholds, adjudication rulings, and scientific interpretations reported in this paper were fixed and made by the human authors, as detailed in Appendix C.

Data availability

The code, pre-specification artifacts, and adjudication trail for this study are available at <https://github.com/jeff-1980/bm3-jizi>. The repository includes all follow-up grids (645 cells with per-epoch curves, including the 210 independent-noise cells of Sections 4.1–4.3), their pre-specifications,

674 drivers, construction checks and decision memos and the harness’s own im-
675 ported modules. The analysis can be recomputed from the released per-
676 cell records alone (it was re-run from a fresh clone and reproduced the
677 archived decisions byte for byte). Retraining requires the XJTU-SY and
678 Paderborn data, a GPU and the Mamba-3 source and kernels (upstream
679 commit `e9594ce`); a portable loader resolves the harness’s imports to the
680 repository copies, and a 2-epoch smoke run of the reverse-transition driver
681 from a clean copy of the repository (not the author’s directory tree) loaded
682 those copies and reproduced the evaluation-label hash, but a full retraining
683 from a clean environment has not been performed.

684 **Appendix A. Fairness protocol**

685 *Appendix A.1. Evaluation-set fingerprint*

686 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
687 256 hash of the raw held-out test-label byte array, computed once per run
688 and asserted identical across every arm/condition/seed cell within a grid.
689 It fingerprints *that the same held-out label vector was scored every time*, not
690 model weights or predictions. It is also the split on which the best-epoch rule
691 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
692 five grids is:

693 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

694 This value is recomputed independently at adjudication time (not merely
695 copied from the run log) and cross-checked row-by-row against every grid’s
696 fairness report before any comparison across arms is treated as valid.

697 *Appendix A.2. Per-arm parameter counts*

698 Table A.7 lists every arm’s parameter count, each obtained directly via
699 `sum(p.numel() for p in model.parameters())` on the instantiated model
700 (not estimated), cross-verified against an independently maintained archi-
701 tecture map.

702 *Appendix A.3. Five-batch pre-specification timeline*

703 Table A.8 lists, for each of the five pre-specified grids, the artifact that
704 fixed its adjudication thresholds, that artifact’s creation/modification date,
705 and the corresponding grid’s run-directory start time. In every case the
706 prereg artifact predates the grid it governs.

Table A.7: Per-arm parameter counts for every arm of the study (from the `n_params` field recorded in each cell and the harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
frozen-trap (<code>frozen_ctrapp</code>)	111,906
frozen-rot (<code>frozen_cangle</code>)	104,346
frozen-all (<code>frozen_clti</code>)	103,842
frozen-all-gate (<code>frozen_clti_nogate</code>)	71,074
frozen-all+add (<code>frozen_clti_add</code>)	103,842
frozen+add (<code>bm3_frozen_add</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802
S4D+gate, width-matched (<code>s4d_plus_gate_wm</code>)	198,978
S4D+additive branch (<code>s4d_plus_branch</code>)	198,978

Table A.8: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (mtime 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

Two honest caveats. First, the baseline batch predates the orchestration infrastructure used for the other four batches by five days; its prereg trail rests on the task card’s creation timestamp alone, not on the same automated executor/reviewer machinery documented in [Appendix C](#), and we state this rather than implying a uniform process across all five batches. Second, for the component-ablation grid, two additional JSON files bearing “pre-reg” in their name exist on disk (`secondary_prereg_20260704-0708` and `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start time; both are explicitly marked void in the corresponding decision artifact’s `void_preregs` field, and neither was used to adjudicate any result in this paper. They are the on-disk residue of the first post-hoc criterion substitution described in [Appendix C](#), retained here (rather than deleted) as part of the released adjudication trail.

Appendix B. Confusion matrices and a repeat-run check

Appendix B.1. Full confusion-matrix set

Table [B.9](#) gives the complete confusion-matrix set underlying the per-class analysis of [Appendix D.10](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB), each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These counts were dumped per-sample and re-verified against the reported recall/precision/F1 metrics cell-by-cell before use.

Table B.9: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	-6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	-6 dB	13,267	4,253	3,244	5,476
frozen $-$ gate	clean	16,935	585	5,139	3,581

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
S4D	0 dB	16,092	1,428	5,406	3,314
	−6 dB	13,200	4,320	5,469	3,251
	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

727 Appendix B.2. Repeat-run check

728 Because two independent process runs of an identical configuration are
729 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
730 ductions in the selective-scan and normalisation ops), we measured a process-
731 level noise floor directly: the same arm, condition, and seed (`bm3.frozen`,
732 clean, seed 0, 2 epochs) was run twice in separate processes with identical
733 evaluation fingerprints and identical parameter counts confirmed. The two
734 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

735 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
736 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
737 run, and therefore neither an estimate nor a bound of process variability for
738 full-length runs. It is an order of magnitude below the seed-level variabil-
739 ity observed within the deep-noise band across the curve tables underlying
740 Figures D.2 and D.4 (standard deviations typically 2–9 pp, e.g. BM3-frozen
741 5.7–9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Sec-
742 tion 4 are judged against seed-level variability, not against this single repeat.

743 Appendix C. Automation disclosure and adjudication gates

744 Appendix C.1. Automation disclosure

745 The mechanical stages of the analysis pipeline—applying pre-specified nu-
746 meric thresholds to aggregated results, re-computing aggregates, re-verifying
747 evaluation fingerprints, and checking each decision against its pre-specified
748 rule—were carried out by LLM-based coding agents under an orchestration
749 harness, with every step auditable in the released trail. Manuscript drafting
750 and figure generation were also assisted by such agents (main text, Declara-
751 tion of Generative AI).

752 *Appendix C.2. Incidents*

753 Three incidents occurred in the original grids, each caught before an er-
754 roneous artefact entered a reported number: (1) on the frozen-selectivity
755 grid an automated decision substituted an unspecified criterion for the pre-
756 specified 10 pp threshold; re-applying the original rule gives the reported out-
757 come, “not the primary mechanism”; (2) on the component-ablation grid an
758 automated pass substituted an interval heuristic for the pre-specified thresh-
759 old, and post-hoc rule files were generated; the true pre-specification predated
760 the grid and was re-applied, and the post-hoc files are retained and marked
761 void; (3) on the graft grid a construction stage was falsely marked as passed;
762 the grid was re-executed from scratch under a corrected task card. Devia-
763 tions in the later follow-up grids (for example a bootstrap made deterministic
764 after first use, and rounding corrections) are recorded in the repository’s de-
765 cision memos. Full descriptions, with dates and artefact names, are given in
766 Supplementary Material S1.

767 *Appendix C.3. Human adjudication gates*

768 All numeric adjudication thresholds, every ruling on an incident, the de-
769 cision to report the incidents, and every interpretive claim in Section 5 were
770 fixed or made by the human authors. No agent could overwrite a protected
771 decision artefact, alter a pre-specified threshold or mark its own output as
772 final.

773 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

774 In line with Elsevier’s policy, AI tools are not listed as authors and the
775 human authors take full responsibility for the content. We extend the dis-
776 closure beyond the writing process to the adjudication pipeline because that
777 is where automation could affect the reported conclusions.

778 **Appendix D. Results under the earlier noise protocol**

779 This appendix collects the results obtained while the noise added to win-
780 dow i of the training and the evaluation set was drawn from the same stream
781 (Section 3.4), together with the final-epoch and best-epoch re-runs of the
782 original grids. They are reported in full because they were the pre-specified
783 adjudications of the original design, but they are of lower evidential weight
784 than the independent-noise results of Section 4: no abstract-level claim rests
785 on them alone, thresholds labelled “primary” or “necessary” below refer to

the earlier-protocol grid only, and where an independent-noise counterpart exists (Table 2) the counterpart takes precedence. Section 4.5 summarises what these results add.

Appendix D.1. Freezing the input-dependent Δ , A , B , C projections (earlier protocol)

Table D.10: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Appendix D.2 and Appendix D.9). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3} - \text{frozen})$	$\Delta(\text{frozen} - \text{S4D})$
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+15.5
−2 dB	88.3	87.3	67.5	+0.9	+19.8
−6 dB	83.3	83.1	66.7	+0.2	+16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
−2 dB	95.8	92.3	75.1	+3.5	+17.2
−6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	−0.0	−2.1
−10 dB*	68.1	64.3	52.3	+3.8	+12.0

Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs at most 3.7 pp anywhere in the adjudication band on the point estimates under final-epoch reporting (+3.7, +0.9, +0.2 pp at 0, −2, −6 dB), while the frozen block retains an advantage of +15.5, +19.8, +16.4 pp over S4D (Table D.10, Figure D.2). Five seeds bound these quantities loosely: paired 95% intervals for $\Delta(\text{BM3} - \text{frozen})$ are [−3.8, 11.3], [−4.5, 6.3], [−12.1, 12.6] pp (BM3 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen} - \text{S4D})$ [4.3, 26.8], [12.0, 27.7], [4.1, 28.7] pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robustness to selectivity only if $\Delta(\text{BM3} - \text{frozen}) \geq 10$ pp in deep noise with the frozen arm collapsing toward S4D. It was written for best-epoch reporting,

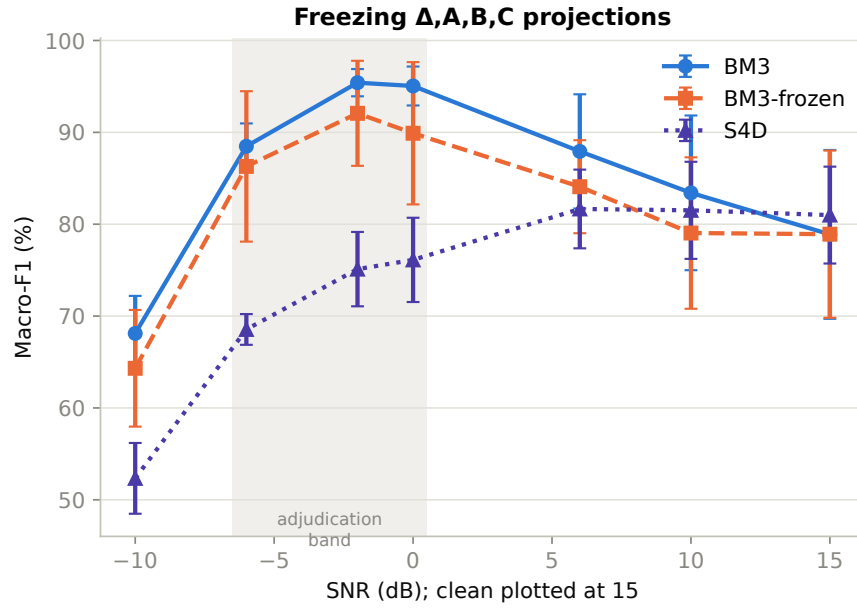


Figure D.2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table D.10, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band. Earlier noise protocol (training and evaluation noise shared by index).

801 under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2$ pp
 802 (re-run: $+5.1, +3.5, +3.0$) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7$ pp.
 803 Neither condition holds under either rule, so the pre-specified verdict is that
 804 these projections are *not the primary mechanism*. This is a failure to meet a
 805 confirmation criterion, not a demonstration that freezing is cheap: the inter-
 806 vals above are compatible with differences larger than 10 pp, and no equiva-
 807 lence margin was pre-specified or tested. One limit applies: the frozen arm
 808 still contains input-dependent trapezoidal weights and rotation angles, so this
 809 result concerns Δ, A, B, C only; [Appendix D.2](#) removes that limit. On clean
 810 data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-
 811 style arms—in line with the clean-benchmark comparison of [4], which, like
 812 this auxiliary row, involves no added noise. In the noisy regime, the block’s
 813 advantage over S4D is not removed by freezing these four terms, so this
 814 freeze does not identify where the advantage comes from; the other parts of
 815 the block and the remaining input-dependent scan terms are candidates.

816 *Appendix D.2. Freezing the remaining input-dependent scan terms (earlier*
 817 *protocol)*

818 Table [D.11](#) reports the three layered arms of Section [3.2](#) on the adjudi-
 819 cation band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original
 820 training code, the same evaluation fingerprint, with the original BM3 arm re-
 821 run in the same grid). The question is whether the advantage that survived
 822 freezing Δ, A, B, C is carried by the two input-dependent terms the frozen
 823 arm retained.

824 It is not. With the scan’s dynamics and read/write coefficients input-
 825 independent, *frozen-all* is $+21.9, +24.0$ and $+21.2$ pp above S4D at 0, -2
 826 and -6 dB under final-epoch reporting, positive in 5/5 seeds at every lev-
 827 el and with intervals excluding zero—if anything a larger margin than the
 828 partially frozen arm’s $+15.5, +19.8, +16.4$ pp (Table [D.19](#)). None of the
 829 three freezing steps shows an accuracy loss in the point estimates: the paired
 830 differences BM3-frozen $-$ frozen-trap, $-$ frozen-rot and $-$ frozen-all are
 831 negative at all three levels (between -1.8 and -6.3 pp under final-epoch re-
 832 porting), i.e. the more heavily frozen arm is numerically ahead, with every in-
 833 terval including zero. The direction is not an artefact of capacity: *frozen-all*
 834 has 103,842 parameters against 112,410 for BM3-frozen and 177,938 for B-
 835 M3. At the deepest level *frozen-all* also exceeds the full BM3 block by
 836 4.5 pp ($[-7.2, -1.9]$ for BM3 $-$ frozen-all, 0/5 seeds in BM3’s favour); we

Table D.11: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

837 report this but do not build on it, since five seeds and one speed pair cannot
838 separate a genuine advantage of a time-invariant scan from a variance effect.

839 Two readings of the whole step are therefore available, and the data do not
840 separate them: input-dependence of the scan contributes nothing measurable
841 here, or it contributes something that the learned constants recover under
842 this training budget. Either way the conclusion for the attribution chain
843 is the same and now rests on a complete intervention rather than a partial
844 one: **the Mamba-style block’s deep-noise advantage over a time-**
845 **invariant S4D does not require any input-dependent term in its**
846 **scan.** What remains to carry it is the rest of the block—the convolutional
847 stem, the heavy-tailed A parameterisation, the normalisations and the output
848 gate—which is what [Appendix D.3–Appendix D.4](#) examine. The re-run also
849 reproduces the original grids: the BM3 arm’s best-epoch means (95.6, 95.8,
850 88.6) exceed the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43
851 and 0.16 pp (a different PyTorch/Triton build, [Appendix D.9](#)).

852 *Appendix D.3. Single-component removal on the frozen base (earlier proto-* 853 *col)*

854 “Necessary” is used here in the operational sense of this ablation: remov-
855 ing the gate from the frozen block, with the other components and the budget
856 unchanged, removes most of the block’s advantage. The gate result is large,

Table D.12: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of [Appendix D.9](#)); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified. Earlier noise protocol (training and evaluation noise shared by index); see [Section 4.1](#) for the independent-noise re-run.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem ($k=7$)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A → S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; intervals include 0

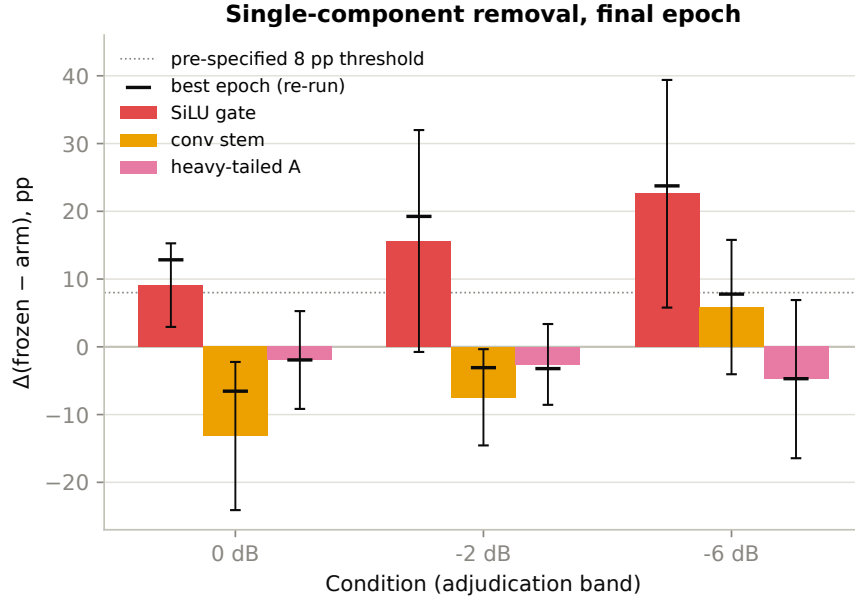


Figure D.3: Component-removal Δ (percentage points, frozen − arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold. Earlier noise protocol (training and evaluation noise shared by index).

one-signed and grows with noise depth: under final-epoch reporting removal
 costs 9.1, 15.6, 22.6 pp at 0, -2 and -6 dB (paired 95% intervals [2.9, 15.3],
 $[-0.8, 32.0]$, [5.8, 39.4]; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp
 threshold at every level; the interval at -2 dB includes zero because of the
 large seed variance of the gateless arm. Under the best-epoch rule for which
 the threshold was written, the component-ablation grid gives 13.1, 21.1 and
 23.5 pp ([6.5, 19.6], [3.8, 38.5], [8.5, 38.6]; 5/5 seeds at every level), so under
 the earlier noise protocol the verdict is *primary* under both rules. Under
 independent training and evaluation noise the same contrast is resolved only
 at -6 dB (Section 4.1), so the operational sense of “necessary” holds there
 at the deepest level only. The gateless arm gives up most of the Mamba
 advantage but is not resolved against plain S4D: frozen-gate - S4D is +6.4,
 +4.2, -6.2 pp, with every interval including zero. We therefore describe it
 as roughly at the level of S4D, below it only on the point estimate at -6 dB;
 this suggests, but does not establish, that components interact rather than
 add. The other two components tell regime stories rather than robustness
 stories. Removing the convolutional stem is beneficial at 0 and -2 dB (frozen
 - frozen-conv = -13.2 [$-24.1, -2.2$] and -7.5 [$-14.6, -0.3$], with the stem-
 less arm ahead in every seed) and costly at -6 dB (+5.9, [$-4.0, 15.8$], 3/5
 seeds); under best-epoch reporting the pattern was the same ($-6.6, -2.7$,
 +7.9 pp). On clean cross-speed data (auxiliary, best epoch, original grid)
 the stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing
 from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pat-
 terns mistransferring across conditions, or with optimisation friction under
 the matched budget; we report both readings. Replacing the heavy-tailed
 state parameterisation by an S4D-style one is numerically beneficial at all
 three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under
 best-epoch reporting), but every paired interval includes zero, so we draw no
 directional conclusion from it.

Appendix D.4. Adding a narrow gate branch to S4D (earlier protocol)

The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-S4D)/(frozen - S4D)}$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level
 0.249, 0.341, -0.003 ; seed-resampled 95% interval [0.09, 0.31]); under the
 best-epoch rule for which it was specified, $R = 0.174$ ([0.04, 0.29]). The pre-
 specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and
 $R < 0.3$ (insufficient); both values fall in the last, although the final-epoch in-
 terval reaches the 0.3 boundary. The per-level paired gains of S4D+gate over

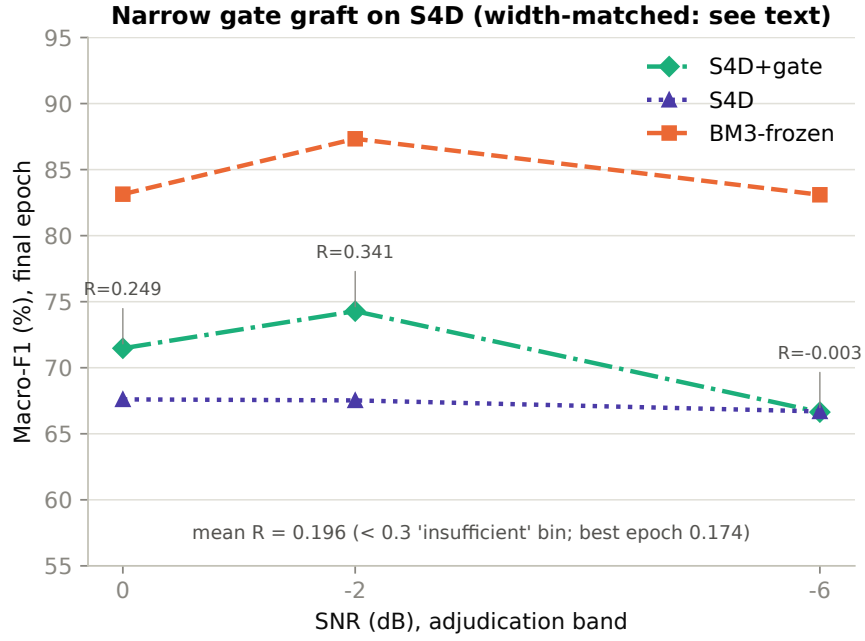


Figure D.4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table D.14. Earlier noise protocol (training and evaluation noise shared by index).

Table D.13: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Appendix D.9). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R , from the component-ablation grid of Appendix D.3 (independent runs of the same arms differ by 0.1–0.7 pp). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

894 S4D under final-epoch reporting are +3.9 ([−0.8, 8.6]), +6.8 ([2.4, 11.1]), −0.1
895 ([−4.0, 3.9]) pp. At the deepest adjudicated level the graft recovers nothing.
896 Removal and addition are not mirror-image interventions. In BM3 the gate
897 has its own $64 \rightarrow 128$ projection and multiplies the scan output at the in-
898 ner width of 128 before the output projection; removing it deletes 32,768
899 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch
900 (with bias) computed from the layer input and multiplying the SSM output
901 before the residual addition; it adds 16,640 parameters over four layers. The
902 two share the multiplicative SiLU form but not width, projection structure
903 or parameter count. Removal costs 9–23 pp in the frozen block, whereas this
904 graft changes S4D by between −0.1 and +6.8 pp (final epoch). What this
905 shows is that *this gate branch, in this host and under this budget, recovers*
906 *little*. Three readings fit the data and cannot be separated here: (i) host de-
907 pendence, i.e. the gate needs the stem or scan context it co-trained with; (ii)
908 implementation differences in width, projection and placement; (iii) trans-
909 plant friction, i.e. the grafted branch is harder to optimise in a foreign host
910 under the matched budget (training converges normally, but convergence is
911 not robustness). Appendix D.5 tests reading (ii) with a width-matched graft
912 and an equal-capacity non-multiplicative control; a tuning or longer-budget

913 comparison selected on a source-domain validation set was not run.

914 The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) -$
915 $F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under
916 last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp
917 at 0, −2 and −6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and
918 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4
919 [−1.4, 30.3] and 23.1 [8.7, 37.4] pp. A positive I means the gate adds more
920 on the frozen host than on S4D. Because the host differs in several respects
921 at once and the two gates are not aligned, I cannot be assigned to a specific
922 component or interaction term, and the seed pairing across arms is nominal
923 (arms share seed indices, not initialisations).

924 *Appendix D.5. A width-matched graft and an additive control (earlier pro-* 925 *TOCOL)*

926 The graft of [Appendix D.4](#) is narrower and smaller than the donor gate.
927 Two further arms rebuild it in the donor’s shape (Table 1). In each layer
928 the layer input is projected to x and z of width 128 (the frozen block’s
929 inner width), the S4D layer runs on x at width 128 with state size 64—so
930 that its own parameter count and total state size equal the host’s 64-wide,
931 state-128 layer—and an output projection maps back to width 64. $S4D+gate$
932 (wm) combines $y \odot \text{SiLU}(z)$, as the donor does; $S4D+branch$ uses the same
933 modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have
934 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938),
935 which biases the test towards recovery. A construction check with forward
936 hooks confirms that the tensor entering the output projection equals the
937 intended formula for each arm and not the other, and that the two arms are
938 identical at initialisation. The thresholds were committed publicly before
939 the grid ran (Section 3.5): R as before, under final-epoch reporting; the
940 verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a
941 bin boundary, in which case we write at the tier less favourable to the claim;
942 and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an
943 interval excluding zero.

944 The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159,
945 0.298, 0.431). Its interval, [0.16, 0.52] under the pre-specified bootstrap
946 ([0.14, 0.72] under paired-seed resampling), crosses the 0.3 boundary, so the
947 pre-specified verdict is INCONCLUSIVE between “insufficient” and “par-
948 tial”, and we report it at the less favourable tier: a graft built in the donor’s

Table D.14: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of $(\text{arm} - \text{S4D})/(\text{BM3-frozen} - \text{S4D})$, with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of [Appendix D.9](#). Earlier noise protocol (training and evaluation noise shared by index); see [Section 4.1](#) for the independent-noise re-run.

Arm	0 dB	−2 dB	−6 dB	R	95% interval
S4D	67.6	67.5	66.7	—	—
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	—
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				−0.112	[−0.26, 0.02]

949 shape recovers part of the gap. The recovery is not gate-specific. The addi-
950 tive branch, with the same parameters, recovers at least as much ($R = 0.408$),
951 $\Delta R = -0.112$ $[-0.26, 0.02]$ (paired-seed $[-0.20, 0.06]$), and at 0 dB the gat-
952 ed arm is below the additive one by 3.0 pp ($[-4.5, -1.5]$, 0/5 seeds). With
953 frozen—all in the denominator the gated arm’s ratio is 0.231 $[0.13, 0.33]$; un-
954 der best-epoch reporting the two ratios are 0.459 and 0.483. What a graft
955 recovers on this transition therefore follows the added structure—a wider
956 channel axis and input/output projections—rather than the multiplicative
957 interaction. Reading (ii) of [Appendix D.4](#) is partly confirmed, in that the
958 narrow graft understated what added structure can recover; but in the S4D
959 host the gate did not contribute measurably beyond an equal-capacity ad-
960 ditive branch, whereas removing it from the frozen Mamba-3 block costs
961 9–23 pp under the same rule ([Table D.19](#)).

962 *Appendix D.6. Replacing the gate in its native host (earlier protocol)*

963 The previous subsection changes the host and keeps the operator. A pre-
964 specified grid (15 cells, original transition) does the converse: it keeps the host
965 and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen
966 layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$;
967 modules, parameter count (112,410) and initial weights are bitwise those of
968 BM3-frozen, and a construction check confirms that the donor operator is
969 $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified
970 statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$,
971 mean over levels, final epoch, with a seed-resampled interval; *not rescued*
972 requires $\rho < 0.3$ with an upper bound below 0.5.

Table D.15: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
	+8.5		+15.8		+22.2	
BM3-frozen − frozen+add	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
	+0.6		−0.2		+0.4	
frozen+add − frozen−gate	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
	+18.1		+22.3		+23.4	
BM3-frozen − frozen+add	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
	+0.3		−0.3		+0.6	
frozen+add − frozen−gate	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

973 The additive branch does not measurably rescue the loss. frozen+add
974 cannot be distinguished from the gateless arm at any level (+0.6, −0.2,
975 +0.4 pp, every interval including zero), and the multiplicative arm exceeds
976 it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070,
977 −0.015, 0.016), with a 95% interval of $[-1.159, 0.486]$ under the pre-specified
978 bootstrap, which resampled each arm and level independently; resampling
979 seeds jointly across arms and levels, as the paired design warrants, gives
980 $[-0.043, 0.136]$. Under best-epoch reporting $\rho = 0.005$. By the pre-specified
981 rule the verdict is *not rescued*: the point-estimate rescue is small and below
982 the pre-specified criterion, which is not a demonstration that the two arms
983 are equivalent.

984 A further pre-specified grid (15 cells) repeats the substitution on the
985 reverse transition (Appendix D.7), with BM3-frozen and frozen−gate taken
986 from the reverse grid. The outcome is the same (Table D.15, lower block):
987 frozen+add cannot be distinguished from the gateless arm at any level (+0.3,
988 −0.3, +0.6 pp, every interval including zero), and the multiplicative arm
989 exceeds it by +18.1, +22.3, +23.4 pp (intervals excluding zero, 5/5 seeds at
990 every level). The rescue ratio is $\rho = 0.009$ (per level 0.017, −0.014, 0.023),
991 with a 95% interval of $[-0.943, 0.362]$ under the pre-specified bootstrap and
992 $[0.002, 0.018]$ under paired-seed resampling; under best-epoch reporting $\rho =$
993 -0.005 . The pre-specified verdict is again *not rescued*.

994 Together with Appendix D.5 this gives the two halves of a symmetric
995 test. In the S4D host the gate did not recover more than an additive branch of
996 the same size on the original transition and recovered somewhat more on the
997 reverse one; in the Mamba-3 host an additive branch of the same size could
998 not be distinguished from removing the gate, on both transitions. Under the
999 earlier noise protocol this pattern is specific to the multiplicative operator
1000 in its native host; Section 4.1 shows that the size of the gate’s contribution
1001 there depends on noise depth.

1002 Appendix D.7. The reverse transition (earlier protocol)

1003 A pre-specified grid repeats the chain’s central contrasts on the reverse
1004 transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen−all, BM3-frozen,
1005 frozen−gate, S4D and S4D+gate (wm), at three levels and five seeds; Sec-
1006 tion 3.1 for the data). The independent-noise re-run of four of these arms
1007 is Section 4.2. This direction is harder for the time-invariant model: on
1008 its evaluation set, predicting the majority class gives a macro-F1 of 46.8%

1009 and uniform random guessing 41.5%, and S4D sits close to those levels (Ta-
1010 ble D.16).

Table D.16: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm (best / final)	0 dB	-2 dB	-6 dB
frozen-all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen-gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen-all - S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen - frozen-gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	-1.8
frozen-gate - S4D	[-17.6, 27.7] (2/5)	[-19.2, 23.9] (2/5)	[-18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) - S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

1011 The first two steps replicate. The fully input-independent scan exceeds
1012 S4D by +26.3, +25.6 and +22.2 pp at 0, -2 and -6 dB, with intervals ex-
1013 cluding zero and 5/5 seeds at every level, so by the pre-specified rule (at
1014 least two of three levels) the first step is replicated on the reverse transition.
1015 Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the
1016 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,
1017 and the gateless arm is indistinguishable from S4D (intervals include zero).
1018 On the third step, which the pre-specification treats as descriptive here, the
1019 width-matched graft recovers most of the gap, $R = 0.911$ [0.82, 1.04] (0.863
1020 with frozen-all in the denominator). A further pre-specified grid (15 cells)
1021 adds the same-size additive control in this direction: it reaches 67.8, 72.1,
1022 68.4% (final epoch) and recovers $R = 0.758$ [0.66, 0.88]. The gate arm ex-
1023 ceeds it by +5.1, +1.7, +3.8 pp (only the -6 dB interval excludes zero),
1024 and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch
1025 reporting it shrinks to 0.037 [-0.1, 0.17]. By the pre-specified descriptive
1026 labels this is *unresolved*: in this direction the gate carries a small increment

over the additive branch, whereas on the original transition it carried none ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available to an additive branch of the same size, and the remaining gate increment is direction-dependent. Both ratios rest on an S4D baseline close to the trivial floor (51–52% against 46.8% for the majority class). Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

Appendix D.8. Paderborn: disjoint and shared bearings (earlier protocol)

A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the writing consequences committed before the first cell) repeats the three central contrasts on the Paderborn KAt data [50], using real-damage bearings only. Models train on two outer-race (KA04, KA16) and two inner-race bearings (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15, KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged; the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms as in XJTU-SY, and everything else (noise, training, arms, budget) is as in the main grids. The pre-specified smoke check (two arms, one extra seed) passed its three criteria, the learnability criterion by 0.3 pp. A second pre-specified grid (45 cells, three arms) then used the same four bearings for training and evaluation, at the same two speeds, from different recordings. This changes the evaluation bearings, their damage profiles, the number of windows and the class balance together, so the two settings are two protocols rather than a single-factor manipulation.

None of the pre-specified contrasts replicates (Table D.17). The fully input-independent block does not exceed S4D (frozen—all – S4D: -1.4 , -2.5 , -3.3 pp, the interval at -6 dB excluding zero in the negative direction), nor does BM3-frozen (-3.8 , -3.6 , -3.4 pp). Removing the gate shows no accuracy cost in this setting: BM3-frozen – frozen–gate is -5.0 , -4.6 , -3.3 pp, below zero in every seed at every level. With no gate effect to rescue, the native-substitution test is not assessable; the additive and gateless arms again cannot be distinguished (0.0 , 0.0 , -0.1 pp). The pre-specified labels are therefore *not replicated* for both freeze contrasts and for gate removal, and *not assessable* for the substitution test.

Two observations bound how this result can be read, and neither changes the labels. First, the testbed discriminates poorly: every arm ends between 47.7 and 56.9% macro-F1 at every level (majority-class predictor 37.5%, a balanced random predictor about 50%), all arms fit the training condition

Table D.17: Paderborn, real-damage bearings, 1500 rpm $\rightarrow$ 900 rpm. *Disjoint*: training on KA04, KA16, KI04, KI14 and evaluation on KA15, KA22, KI17, KI18, KI21. *Shared*: training and evaluation on KA04, KA16, KI04, KI14 (different recordings). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds (majority-class macro-F1: 37.5% disjoint, 33.3% shared). Lower block: paired differences under final-epoch reporting, 95% t -interval and seeds (of 5) with a positive difference. “–”: arm not run in that setting. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm / contrast	Disjoint bearings			Shared bearings		
	0 dB	−2 dB	−6 dB	0 dB	−2 dB	−6 dB
BM3-frozen	51.9	50.3	47.7	84.5	85.8	86.6
frozen—all	54.2	51.3	47.8	—	—	—
S4D	55.6	53.9	51.1	76.1	77.3	77.9
frozen—gate	56.9	54.9	51.1	75.5	78.1	81.3
frozen+add	56.9	54.8	51.0	—	—	—
	−3.8	−3.6	−3.4	+8.4	+8.5	+8.7
BM3-frozen	[−11.9, 4.3]	[−12.6, 5.4]	[−8.9, 2.1]	[−1.3, 18.1]	[0.2, 16.8]	[4.3, 13.0]
– S4D	(1/5)	(2/5)	(1/5)	(4/5)	(5/5)	(5/5)
	−5.0	−4.6	−3.3	+9.0	+7.7	+5.3
BM3-frozen	[−8.8, −1.3]	[−9.2, 0.1]	[−7.1, 0.4]	[5.2, 12.8]	[5.0, 10.4]	[2.4, 8.2]
– frozen—gate	(0/5)	(0/5)	(0/5)	(5/5)	(5/5)	(5/5)
	−1.4	−2.5	−3.3			
frozen—all	[−4.0, 1.2]	[−5.3, 0.2]	[−5.8, −0.8]			
– S4D	(1/5)	(1/5)	(0/5)	—	—	—
	0.0	0.0	−0.1			
frozen+add	[−0.9, 0.8]	[−0.4, 0.3]	[−0.4, 0.2]			
– frozen—gate	(2/5)	(3/5)	(1/5)	—	—	—

(the gated arms reach the lowest training loss), and none transfers far above the level reached after the first epoch. Second, where differences are resolved they run against the Mamba-3 arms, which reach lower training loss and transfer slightly worse. In the disjoint setting, therefore, the block’s advantage under matched noise and drift is not observed; we do not know whether this reflects a property of the PU transition (speed reduction to 900 rpm with disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a limit of the XJTU-SY result itself, which the shared-bearing setting below separates.

Shared bearings. With bearing identity shared between training and evaluation, the same condition pair becomes transferable: every arm ends between 75.5 and 86.6% at every level, and the pre-specified transferability gate (5-seed mean $\geq 65\%$ with one-sided 95% lower bound $> 60\%$ at two of three levels for at least one arm) is passed by all three arms at all three levels (smallest lower bound 71.3%). Within this setting both effects replicate under the pre-specified rule (Table D.17, right). BM3-frozen exceeds S4D by +8.4, +8.5, +8.7 pp at 0, -2 , -6 dB, with intervals excluding zero at -2 and -6 dB; removing the gate costs +9.0, +7.7, +5.3 pp, with intervals excluding zero at all three levels and in every seed; no level runs in the opposite direction. The gateless arm returns to the level of S4D (frozen–gate – S4D: -0.6 , $+0.8$, $+3.4$ pp, all intervals including zero), as on XJTU-SY. Under best-epoch reporting the freeze contrast is resolved at all three levels (+6.9, +10.8, +13.7 pp) and gate removal only at -6 dB (+1.5, +3.7, +6.4 pp). The cost of gate removal reaches the 8 pp threshold used on XJTU-SY at one level only; we compare directions, not magnitudes, across datasets.

Read together, the two protocols give different results: with shared bearings every arm, S4D included, ends 18.6–38.8 pp higher than with disjoint bearings (independent seeds; per-arm Welch intervals all exclude zero), and the block’s advantage and the cost of gate removal reappear. Because the evaluation bearings, class balance and sample size change together, this does not isolate bearing overlap as the cause. With two bearings per class in the shared setting, a bearing’s identity also determines its label, so a model can succeed partly by recognising bearing-specific signatures. XJTU-SY differs from both Paderborn settings: its training and evaluation bearings are disjoint in each direction, and the block transfers there. The shared-setting contrasts were re-run under independent noise (Section 4.1); *frozen–all* was not run on Paderborn.

Appendix D.9. Sensitivity to the reporting rule (earlier protocol)

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms BM3-frozen, S4D, frozen-gate and S4D+gate (60 cells), the BM3 arm alongside the layered freezes (Appendix D.2), and the capacity control and the two other single-component removals, S4D-wide, frozen-conv and frozen-A (45 cells, pre-specified with a rule for changes of sign). The re-runs used a different PyTorch/Triton build from the original grids; their best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp across the eight re-run arms (the largest deviation is in the high-variance frozen-gate arm at -2 dB).

Table D.18: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		-2 dB		-6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen-gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen-conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen-A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Best-epoch values exceed final-epoch values by 0.6 – 10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0 , -2 , -6 dB), and the contrasts that carry the argument are correspondingly stable (Table D.19).

Table D.19: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

1121 *Selectivity.* The frozen block’s advantage over S4D is +15.5, +19.8 and
1122 +16.4 pp (and +21.9, +24.0, +21.2 pp for the fully frozen scan, [Appendix](#)
1123 [D.2](#)), versus +14.6, +17.2 and +17.1 pp under best-epoch reporting in the
1124 re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding
1125 zero at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp
1126 (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp
1127 pre-specified threshold at every level; the interval at −2 dB includes zero
1128 ([−0.8, 32.0] pp, 4/5 seeds) because of the large seed variance of the gateless
1129 arm (standard deviation 14.6 pp at −2 dB). *Gateless versus S4D.* Under final-
1130 epoch reporting the gateless arm is not below S4D at 0 and −2 dB (point
1131 estimates +6.4 and +4.2 pp) and is below it at −6 dB (−6.2 pp); all three
1132 intervals include zero. The statement that gate removal “falls below S4D”
1133 is therefore supported only as a point-estimate observation at the deepest
1134 level, and we phrase it as “to or below” elsewhere. *Graft.* The recovery
1135 ratio is 0.249, 0.341 and −0.003 per level and $R = 0.196$ on average (best-
1136 epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point
1137 estimate. The seed-resampled 95% interval is [0.09, 0.31] and so reaches the
1138 0.3 boundary: with five seeds the data support “far below confirmation (0.7)
1139 and probably below 0.3”, not a precise value of R . The graft gain over S4D
1140 is resolved at −2 dB only (+6.8 pp, [2.4, 11.1]) and is indistinguishable from
1141 zero at −6 dB.

1142 *Remaining arms.* By the rule pre-specified for the 45-cell grid, no arm
1143 changes sign between the two rules at any level where both values exceed
1144 1 pp, so all main-table entries are reported under the final-epoch rule with

1145 unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-
 1146 wide by +15.9, +20.8, +16.0 pp (intervals excluding zero at all three levels),
 1147 and S4D-wide – S4D is –0.4, –1.0, +0.4 pp (intervals including zero). For
 1148 the stem the final-epoch rule sharpens the regime trade-off (intervals at 0
 1149 and –2 dB now exclude zero) without reversing it.

1150 Because the final epoch of the same evaluation condition has been in-
 1151 spected repeatedly during this study, last-epoch values are also not a fresh
 1152 confirmatory test. The reporting rule therefore changes absolute levels and
 1153 the strength of the gateless-versus-S4D statement, but not the ordering BM3-
 1154 frozen > {S4D+gate, frozen–gate, S4D}, the size of the frozen block’s ad-
 1155 vantage over S4D, or the verdicts of the pre-specified gate and graft rules.

1156 *Appendix D.10. Per-class recall (earlier protocol)*

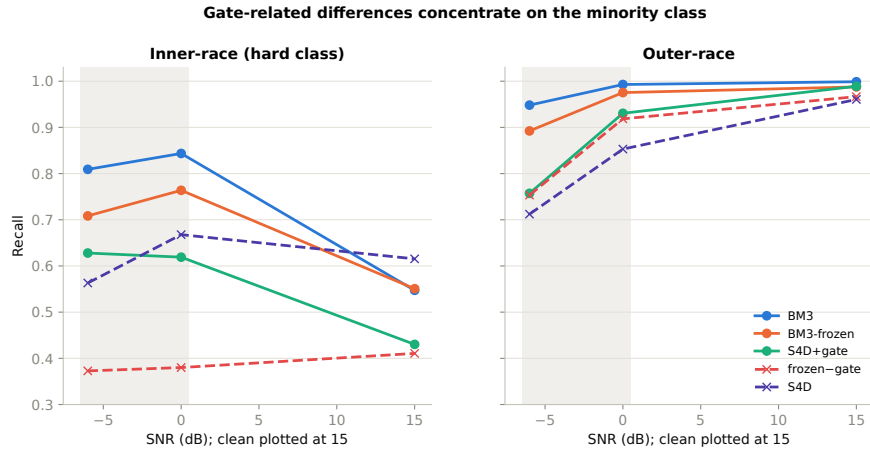


Figure D.5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate. Earlier noise protocol (training and evaluation noise shared by index).

1157 Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch
 1158 selection on the evaluation condition: 5 arms $\times$ {clean, 0, –6 dB} $\times$ 5 seeds,
 1159 per-sample predictions dumped and re-verified against reported metrics cell-
 1160 by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62
 1161 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate
 1162 presence: in all three gated arms IR recall at –6 dB exceeds its clean
 1163 (e.g., 0.55 $\rightarrow$ 0.81 for BM3), while in both gateless arms it is below its clean

value (frozen—gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone, rising to 0.67 at 0 dB) (Figure D.5). Both ablation directions hit IR hardest—gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from predictions pooled over seeds), and the graft’s noise-time gains also concentrate on IR (−6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages the two classes, the arm-level robustness gaps of Appendix D.1–Appendix D.4 are carried substantially by the hard class. The gate-related differences are therefore class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a “low-margin” class. Because this grid is separate from the main 50-epoch grid and the IR class comes from a single training bearing (Section 3.1), these results are descriptive and do not decompose the main-grid effects.

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